

Mini Project Week 5

Team: Jianing Bao, Manu Daza, Martin Grillo, Bernardo Ortega

Date: 2026 - 07 - 17 (Fri.)

CONTENT

Dataset [king-country-houses-aa](#):

Contains **20** house **features** plus the price in **21613** observations.

- data observation
- model baselines
- feature engineering
- model hyper parameter decision
- obstacles
- findings - most gaining features + limits for prediction

DATA OBSERVATION

20 + 1 columns, 21613 rows

no duplicates, no Null values

'id' → drop()

'date' - object → convert #all others: numerical

outlier: 33 '**bedrooms**' (0 'bathrooms', 0 'bedrooms')

'zipcode' - large values could overestimate importance...

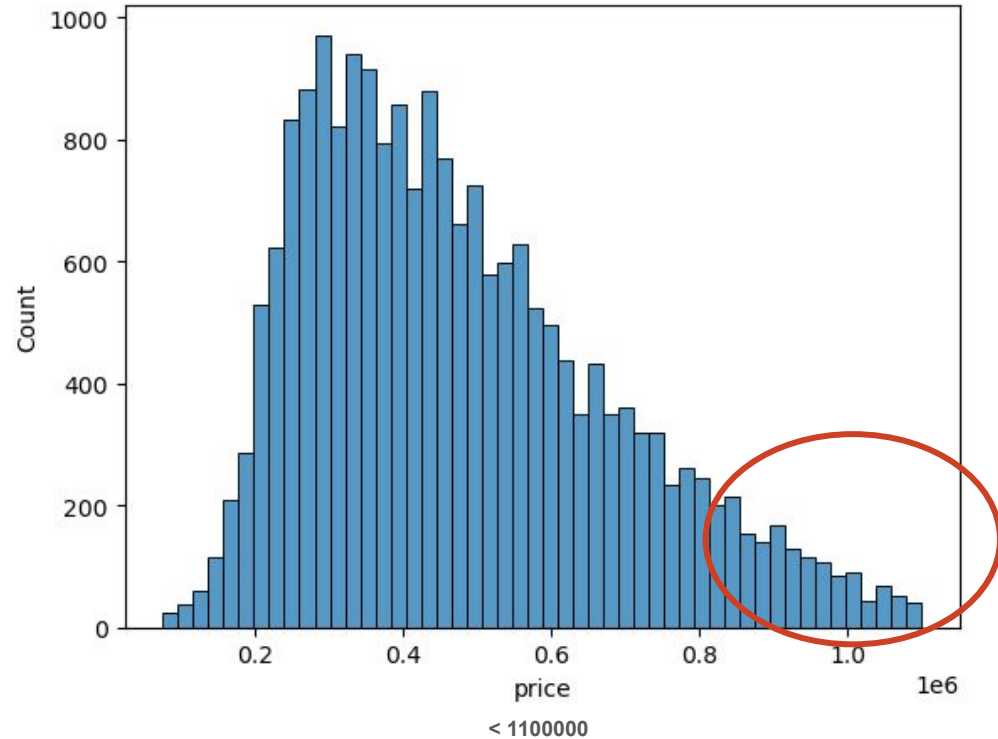
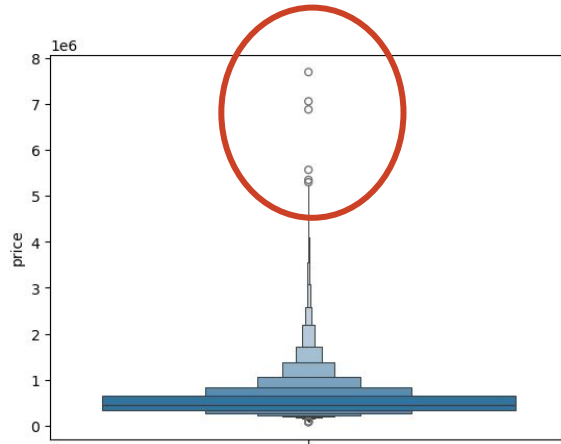
years (as in 'yr_built', 'yr_renovated') - large values → convert to age

DATA OBSERVATION

'price' - right skewed

IQR: upper_bound = 1.129.575

1140 houses more expensive



MODEL BASELINES

Regression model comparison

Name	train pred - R^2 score	test pred - R^2 score	Comment
XGBRegressor	0.98	0.86	overfitted
DecisionTree	0.99	0.73	overfitted
Linear Regression	0.70	0.70	not overfitted
KNN (K=5)	0.87	0.78	overfitted
Random Forest	0.98	0.88	overfitted

PREPROCESSING & FEATURE ENGINEERING

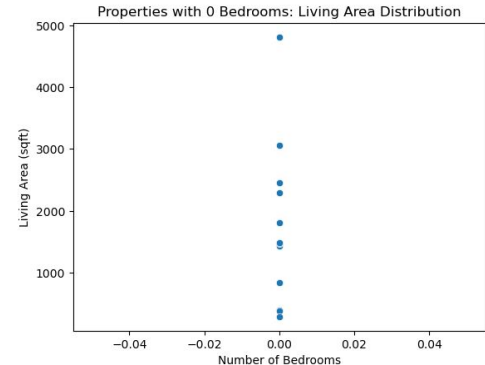
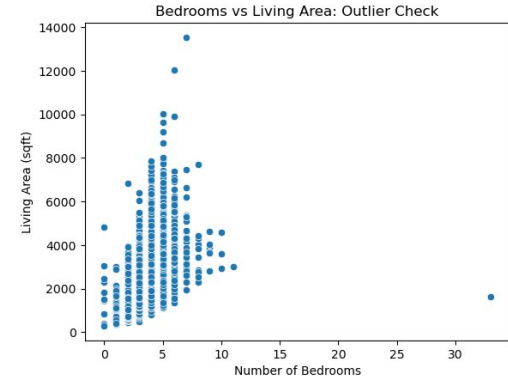
drop: 'bedrooms' > 32 & 'bedrooms' = 'bathrooms' = 0

years -

df['sale_year'] = pd.to_datetime(df['date']).dt.year

df['age_at_sale'] = 'sale_year' - 'yr_built'

'zipcode' – OneHotEncoder



PREPROCESSING & FEATURE ENGINEERING

drop: 'bedrooms' > 32 & 'bedrooms' = 'bathrooms' = 0

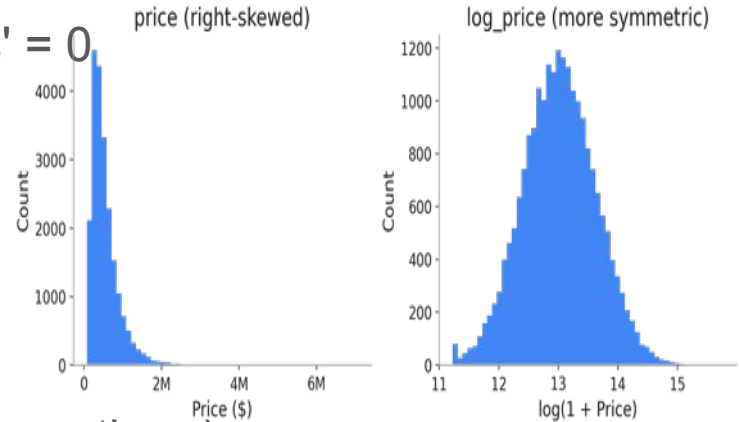
years -

df['sale_year'] = pd.to_datetime(df['date']).dt.year

df['age_at_sale'] = 'sale_year' - 'yr_built'

'zipcode' – OneHotEncoder (strd + rob Scaler on others)

df['log_price'] = np.log1p(df['price'])



PREPROCESSING & FEATURE ENGINEERING

- outliers in price have to be sealed - **log('price')** is best treatment
- optimization on bed-&bathrooms, date (years) been ineffective

transforming to 'age_at_sale', 'sale_year', 'living_to_lot_ratio', 'dist_to_seattle'
drop outliers in rooms

- standardization (scaling for symmetrical feat) been ineffective
- robust scaling (feat heavily skewed, outliers) been ineffective
- OneHotEncoder on 'zipcode' been ineffective

MODEL HYPER PARAMETER DECISION

RandomizedSearchCV

```
param_grid = {learning_rate,n_estimators,subsample,min_child_weight,gamma,reg_lambda (L2),reg_alpha(L1)}
```

```
output =      {-0.2,    +200,   -0.3,    +6,           0.1,    +4,           0.1}
```

more robust - hasn't been effective on score

test pred XGBRegressor (r^2 score): 0.9

OBSTACLES

Motivation - after finding a very good model + log('price')

measurements did not show any improvements for standardScaling,
HotOneEncoding or Model Optimization.

Findings:

- Applying **log** on 'price' was most effective data processing action
- good default models are hard to improve
- Hyperparameter: Manual vs. CrossValidation?

Manually:	R2 test is 0.7526,	R2 train is 0.8229,	diff is 0.0703
CrossValidation:	R2 test is 0.7759,	R2 train is 0.9277,	diff is 0.1518

Findings:

- Hyperparameter: **Manual** vs. **CrossValidation**?

-- **GridSearchCV** optimizes the average cross-validation performance within the predefined parameter grid. Reliable.

– **Manual tuning** (here: for loops) allowed us to further explore parameter combinations and choose a model with better performance on our evaluation metrics.

→ Final model selection:

based on the best performance between **Test R^2** and generalization (**small train-test gap**).

Model: decision Tree

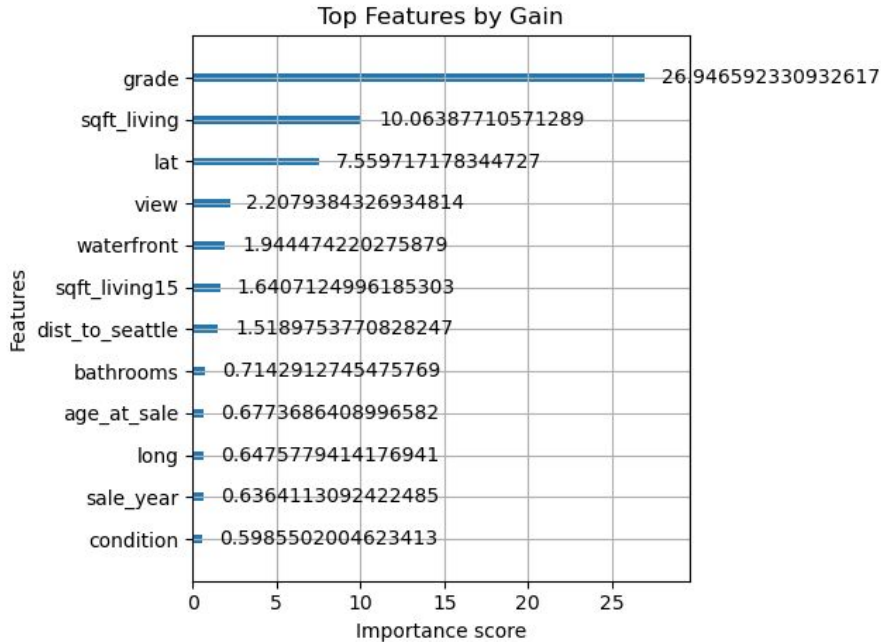
MODEL OPTIMIZED

Regression model comparison

Name	train pred - R^2 score	test pred - R^2 score
XGBRegressor	0.98 $\rightarrow$ 0.94	0.86 $\rightarrow$ 0.90
DecisionTree	0.99 $\rightarrow$ 0.83	0.73 $\rightarrow$ 0.82
Linear Regression	0.70 $\rightarrow$ 0.82	0.70 $\rightarrow$ 0.82
KNN (K=10)	0.89 $\rightarrow$ 0.86	0.81 $\rightarrow$ 0.81
Random Forest	0.98 $\rightarrow$ 0.98	0.882 $\rightarrow$ 0.887

findings - most gaining features + limits for prediction

comparing resulting feature set with corr to 'price'



STR. SPEARMAN CORR		
grade	:	0.658
sqft_living	:	0.644
sqft_living15	:	0.572
sqft_above	:	0.542
bathrooms	:	0.497
lat	:	0.456
bedrooms	:	0.345
floors	:	0.322
view	:	0.294
sqft_basement	:	0.252
waterfront	:	0.115
yr_built	:	0.102
yr_renovated	:	0.102
sqft_lot	:	0.075

THANKS